

Practicum No.8**Date:.....****DMA Simulation****I. Practical Significance**

This practicum helps students understand how data can move quickly between memory and devices without slowing down the CPU. It shows the importance of DMA in real systems where large files, videos, or sensor data must be transferred efficiently. By simulating DMA, students can see how the CPU stays free for other tasks while the transfer continues in the background. This knowledge connects classroom concepts to real-world uses like gaming, streaming, and embedded systems where speed and multitasking are critical.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency.

- **Evaluate** the performance of scalable architectures for intelligent systems to ensure reliability.
(As stated in the Course Structure)
- a. **Implement** cache mapping techniques and simulate direct memory access operations. (PDO)

III. Relevant CO(s) (As stated in the Course Structure)

- CO3. **Interpret** memory functions and techniques influencing performance.

IV. Practical Outcome (PrO) (As stated in the Course Structure)

- **Simulate** DMA operations for **the given** memory setup using MATLAB

V. Related ADO(s) (As stated in the Course Structure)

- **Follow** industry standards for system evaluation and performance optimization.
- **Function** effectively in teams to develop processor and memory architectures.
- **Function** responsibly in leadership roles to design computing systems.

VI. Minimum Underpinning Theory (Include relevant content & images for this practicum.)

Direct Memory Access (DMA) is a technique in computer systems that allows data to be transferred directly between an input/output (I/O) device and the main memory without continuous involvement of the Central Processing Unit (CPU). Normally, the CPU manages all data movement, which can slow down processing, but with a DMA controller, data transfers are handled independently, allowing the CPU to focus on other operations.

This improves system efficiency and throughput, especially in applications where large amounts of data must be moved quickly, such as multimedia, disk operations, or real-time communication. By bypassing the CPU and reducing bottlenecks, DMA ensures faster and smoother performance, making it a vital concept in modern computer architecture.

VII. Algorithm (Include probable photos of experimental set-up.)

1. **Start** the program.
2. **Initialize** memory as an array (example: memory = zeros(1,20) to store data).
3. **Initialize** I/O device data (example: deviceData = randi([1 99],1,5)).
4. **Display** CPU idle message (to show CPU is not moving data).
5. **Transfer** data directly from deviceData to specific locations in memory using DMA logic (example: memory(1:5) = deviceData).
6. **Print** updated memory contents to show DMA transfer.
7. **Print** CPU free message (to indicate CPU was bypassed).
8. **End** the program.

VIII. Resources Required

S. No.	Resource	Suggested Broad Specification	Quantity
1	Laptop	Intel Core i5/i7 processor, 8GB RAM, Windows OS recommended for 8086 emulator	1 No.
2	MATLAB	https://matlab.mathworks.com/	1 No.

IX. Safety Precautions

- Do not run unfamiliar or unverified scripts found online which may harm your computer.
- Verify your register usage and memory addresses to avoid system-level crashes or infinite loops.
- Close all other programs before executing large or unknown assembly code to free up resources.
- Use licensed version of MATLAB to avoid security or legal issues.

X. Procedure (*Self-Instructional*)

- Switch on the computer.
- Open MATLAB on your computer.
- Create a new script file (File → New → Script).
- Run the following MATLAB script and note the values on the console.

% Practicum: Illustrate DMA Operations

clc; clear;

% Step 1: Initialize memory (20 locations, all zero)

memory = zeros(1,20);

% Step 2: Generate random data from I/O device (5 values)

deviceData = randi([1 99],1,5);

disp('Initial Memory:');

disp(memory);

% Step 3: Show CPU is free (not moving data)

disp('CPU is free. DMA will handle data transfer...');

% Step 4: DMA transfers data directly into memory (locations 1 to 5)

memory(1:5) = deviceData;

% Step 5: Display updated memory contents

disp('Updated Memory after DMA Transfer:');

disp(memory);

% Step 6: Show data received from I/O device

disp('Data received from device:');

disp(deviceData);

% Step 7: Indicate completion

disp('DMA transfer completed without CPU involvement.');

- Shutdown the computer

```

Initial Memory:
Columns 1 through 14

    0    0    0    0    0    0    0    0    0    0    0    0    0    0

Columns 15 through 20

    0    0    0    0    0    0

CPU is free. DMA will handle data transfer...
Updated Memory after DMA Transfer:
Columns 1 through 14

   81   90   13   91   63    0    0    0    0    0    0    0    0    0

Columns 15 through 20

    0    0    0    0    0    0

Data received from device:
   81   90   13   91   63

```

deviceData	[81,90,13,91,63]	1×5	double
memory	1×20 double	1×20	double

ADDITION QUESTION:

1. Evaluate how DMA improves system performance compared to CPU-controlled data transfer.

CODE:

```

clc; clear;
close all;

```

```

dataSize = 1000000;
data = rand(1, dataSize);
tic

```

```

memory_cpu = zeros(1, dataSize);
for i = 1:dataSize
    memory_cpu(i) = data(i);
end
cpu_time = toc;

```

```

tic
memory_dma = data;
dma_time = toc;

```

```

fprintf('CPU Controlled Transfer Time: %f seconds\n', cpu_time);
fprintf('DMA Transfer Time: %f seconds\n', dma_time);

```

```

if dma_time < cpu_time
    disp('DMA transfer is faster and improves system performance.');
```

```

else
    disp('CPU transfer took similar or less time.');
```

end

```
CPU Controlled Transfer Time: 0.007320 seconds
DMA Transfer Time: 0.000065 seconds
DMA transfer is faster and improves system performance.
```

cpu_time	0.0073	1×1	double
data	1×1000000 d...	1×1000000	double
dataSize	1000000	1×1	double
dma_time	6.4900e-05	1×1	double
i	1000000	1×1	double
memory_...	1×1000000 d...	1×1000000	double
memory_dma	1×1000000 d...	1×1000000	double

2. Demonstrate how bypassing the CPU during data transfer saves time and power.

CODE:

```
clc; clear;
close all;
```

```
dataSize = 500000;
```

```
data = rand(1,dataSize);
```

```
cpuMemory = zeros(1,dataSize);
tic
```

```
for i = 1:dataSize
    cpuMemory(i) = data(i);
end
```

```
cpuTime = toc;
```

```
tic
dmaMemory = data;
dmaTime = toc;
```

```
cpuPower = cpuTime * 50;
dmaPower = dmaTime * 50;
```

```
fprintf('CPU Controlled Transfer Time: %f seconds\n', cpuTime);
fprintf('DMA Transfer Time: %f seconds\n', dmaTime);
fprintf('Estimated CPU Power Consumption: %f units\n', cpuPower);
fprintf('Estimated DMA Power Consumption: %f units\n', dmaPower);
```

```
if dmaTime < cpuTime
    disp('Bypassing CPU using DMA saves time.');
```

end

```
if dmaPower < cpuPower
disp('DMA also reduces power consumption.');
```

end

cpuMem...	1×500000 do...	1×500000	double
cpuPower	0.2982	1×1	double
cpuTime	0.0060	1×1	double
data	1×500000 do...	1×500000	double
dataSize	500000	1×1	double
dmaMem...	1×500000 do...	1×500000	double
dmaPower	0.0061	1×1	double
dmaTime	1.2250e-04	1×1	double
i	500000	1×1	double

```
CPU Controlled Transfer Time: 0.005964 seconds
DMA Transfer Time: 0.000122 seconds
Estimated CPU Power Consumption: 0.298180 units
Estimated DMA Power Consumption: 0.006125 units
Bypassing CPU using DMA saves time.
DMA also reduces power consumption.
```

XI. Actual Resources Used *(To be filled by the students)*

S. No.	Name of Resource	Broad Specifications		Data	Remarks (If any)
		Tool	Version		
1.	Laptop	Intel Core i5 Processor, 8GB RAM	MATLAB	R2023a	Used for simulation
2.	MATLAB Software	Numerical computing and simulation software	MATLAB	R2023a	Used to run DMA program
3.	Internet	Broadband connection	Chrome	Latest	Used for reference
4.	Operating System	Windows OS	Windows	Windows 11	System environment

XII. Actual Procedure Followed

1. The computer system was switched on and MATLAB software was opened.
2. The provided MATLAB program for **DMA simulation** was typed in the script editor.
3. Memory was initialized as an array with **20 locations** using `memory = zeros(1,20)`.
4. Random data from the I/O device was generated using `deviceData = randi([1 99],1,5)`.
5. The CPU idle message was displayed to indicate that the CPU is not involved in data transfer.
6. The DMA controller transferred the device data directly to memory locations **1 to 5**.

XIII. Observations and Dimensional Measurement

S. No.	Observations	Data	Remarks
1	Memory Size	20 locations	Initialized using zeros
2	Device Data Size	5 values	Generated randomly
3	Data Transfer Method	Direct Memory Access	CPU not involved
4	Memory Update	First 5 locations updated	Successful DMA transfer

XIV. Results/Observations

The MATLAB program successfully simulated the **Direct Memory Access (DMA) operation**. Data generated from the I/O device was transferred directly into memory locations without CPU intervention. The updated memory array confirmed that the DMA controller handled the data transfer efficiently.

XV. Interpretation of Results

The results show that DMA allows data transfer directly between an I/O device and memory without continuous CPU involvement. This improves system performance because the CPU remains free to perform other tasks while the DMA controller handles the data transfer. This technique is widely used in systems that require fast and efficient data movement.

XVI. Conclusions

The practicum successfully demonstrated the **working principle of Direct Memory Access using MATLAB simulation**. The experiment showed how DMA transfers data directly from an I/O device to memory while keeping the CPU free for other processes. This technique significantly improves system efficiency and is commonly used in modern computing systems.

XVII. Suggested Practicum Related Questions:

Note: Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes.

1. **Explain** why industries prefer DMA instead of letting the CPU handle all data transfers.
2. **Illustrate** how DMA makes multitasking (like browsing while playing music) possible.
3. **Predict** what could happen if DMA was not used in a high-speed device like a hard disk.
4. **Evaluate** how DMA improves system performance compared to CPU-controlled data transfer.
5. **Assess** why DMA is important in real-time systems like robotics or medical devices.
6. **Demonstrate** how bypassing the CPU during data transfer saves time and power.
7. **Examine** the role of a DMA controller in efficient memory management.
8. **Discuss** how frequent CPU involvement in data transfer might affect online gaming.
9. **Justify** why DMA is critical for large file transfers in industries like video editing or banking.
10. **Propose** how understanding DMA can help engineers design faster computer systems.

11.

XVIII. References / Suggestions for further reading

- a) I/O Interface (Interrupt and DMA Mode) - <https://www.geeksforgeeks.org/computer-organization-architecture/io-interface-interrupt-dma-mode/>
 b) Interrupts and DMA - <https://youtu.be/2W5igWiI-8>

XIX. Suggested Assessment Scheme

The performance indicators given serve as a guideline for assessing the 'process-related skills/LOs' (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and 'product-related skills/LOs'

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the competency related to the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related Skills: 18 Marks - 60%			
1	<u>Follow</u> industry standards	3	
2	<u>Function</u> effectively in teams	2	
3	<u>Function</u> responsibly in leadership	3	
4	<u>Implement</u> cache mapping techniques and simulate direct memory access operations	10	
Product-related Skills: 12 Marks - 40%			
5	Uses correct commands or syntax in Tool	3	
6	Results	3	
7	Interpretation of Results	3	
8	Conclusion	3	
Total		30	

To be filled by Student:

S.No.	Name of the Student	Register No.
1	NEAVIS RISHVA. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills (18)	Product-Related Skills (12)	Total (30)		

